Practice workbook · Classify immunity as natural or artificial and active or passive

Distinguishing natural, artificial, active and passive immunity, including antibodies from mothers or injected donor plasma.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-CLASSIFY-IMMUNITY-TYPES

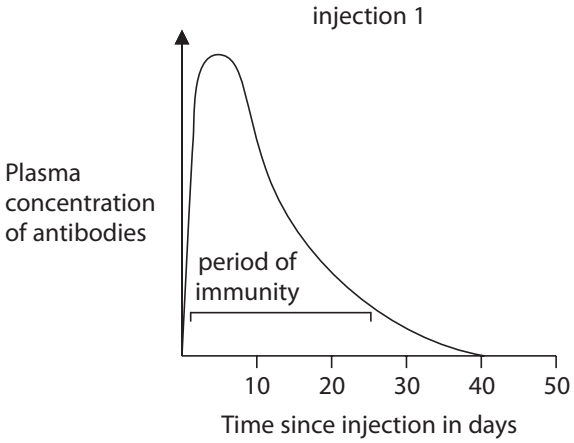
1. Question 4 · part a.i · 2 marks

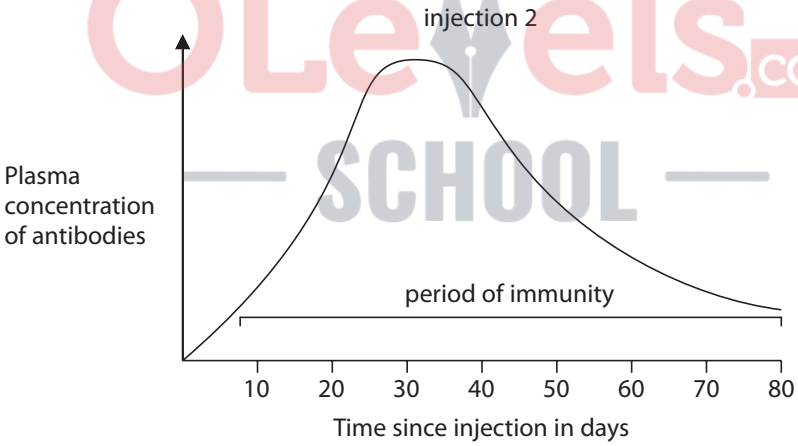
January 2022 Paper 2

Worked-example retry · Attempt part a.i only.

4 (a) Artificial immunity can be produced by different types of injection.

The graphs show the plasma concentration of antibodies produced as a result of two types of injection.





(i) Identify the different forms of artificial immunity produced by these two injections.

(2)

1 .....

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1. Question 4 · part a.i · 2 marks

January 2022 Paper 2

Worked-example retry · Attempt part a.i only.

(ii) Explain which injection would be better for someone who already has the disease.

(3)

(b) Describe how natural immunity to a disease can happen.

(5)

(Total for Question 4 = 10 marks)

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1. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a.v · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.v only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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2. Question 2 · part a.v · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.v only.

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(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

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(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

2. Question 2 · part a.v · 2 marks

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Further practice · Attempt part a.v only.

(iv) Suggest why donated CWB should never be frozen (line 17).

(2)

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

(2)

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

(3)

(Total for Question 2 = 14 marks)

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2. Question 2 · part a.v · 2 marks · Working and answer

Working and answer

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3. Question 7 · part b · 6 marks

November 2020 Paper 2

Further practice · Attempt part b only.

7 (a) Give two differences between whole blood and plasma.

(2)

1 .....

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2 .....

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(b) Ebola is an infectious disease. One symptom of Ebola is blood loss.

The World Health Organisation (WHO) has a procedure for treating Ebola.

This is the WHO's procedure.

- find a person who has recovered from Ebola
- observe this person to make sure that they are disease free for 28 days
- take blood from this person and test the blood for other diseases
- separate the plasma from the blood
- transfer this plasma to another person who has Ebola

Explain how the WHO procedure helps to treat a person who has Ebola.

(6)

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3. Question 7 · part b · 6 marks

November 2020 Paper 2

Further practice · Attempt part b only.

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(Total for Question 7 = 8 marks)

Working and answer

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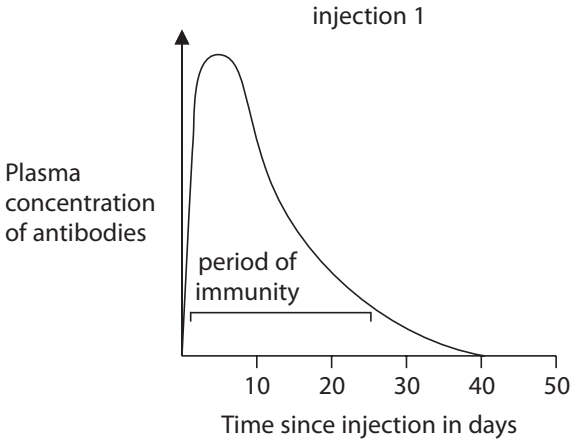
4. Question 4 · part a.ii · 3 marks

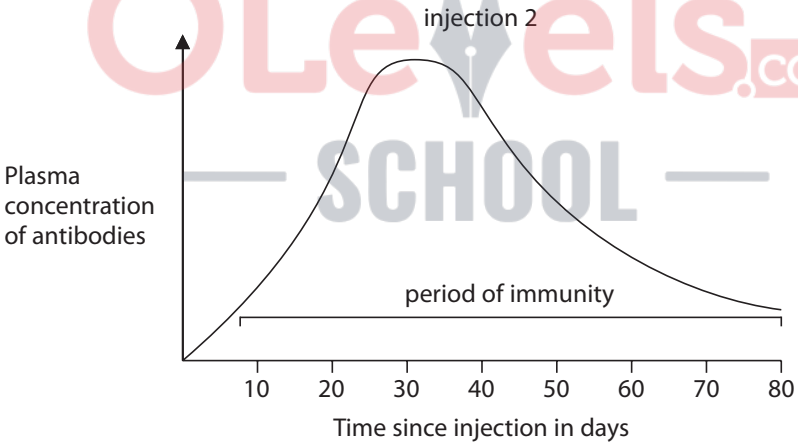
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Further practice · Attempt part a.ii only.

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4. Question 4 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

(ii) Explain which injection would be better for someone who already has the disease.

(3)

(b) Describe how natural immunity to a disease can happen.

(5)

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4. Question 4 · part a.ii · 3 marks · Working and answer

Working and answer

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5. Question 3 · part b.ii · 2 marks

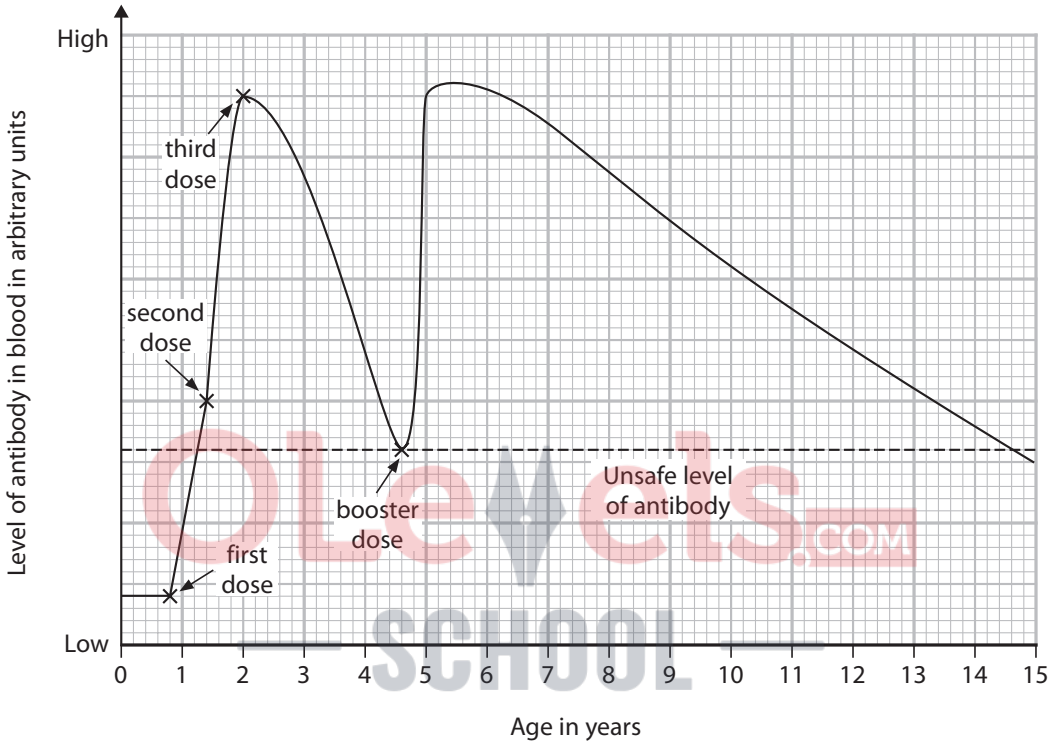
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Further practice · Attempt part b.ii only.

3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

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5. Question 3 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

(2)

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2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

(2)

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(d) Explain why a booster dose is necessary.

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5. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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6. Question 3 · part c · 2 marks

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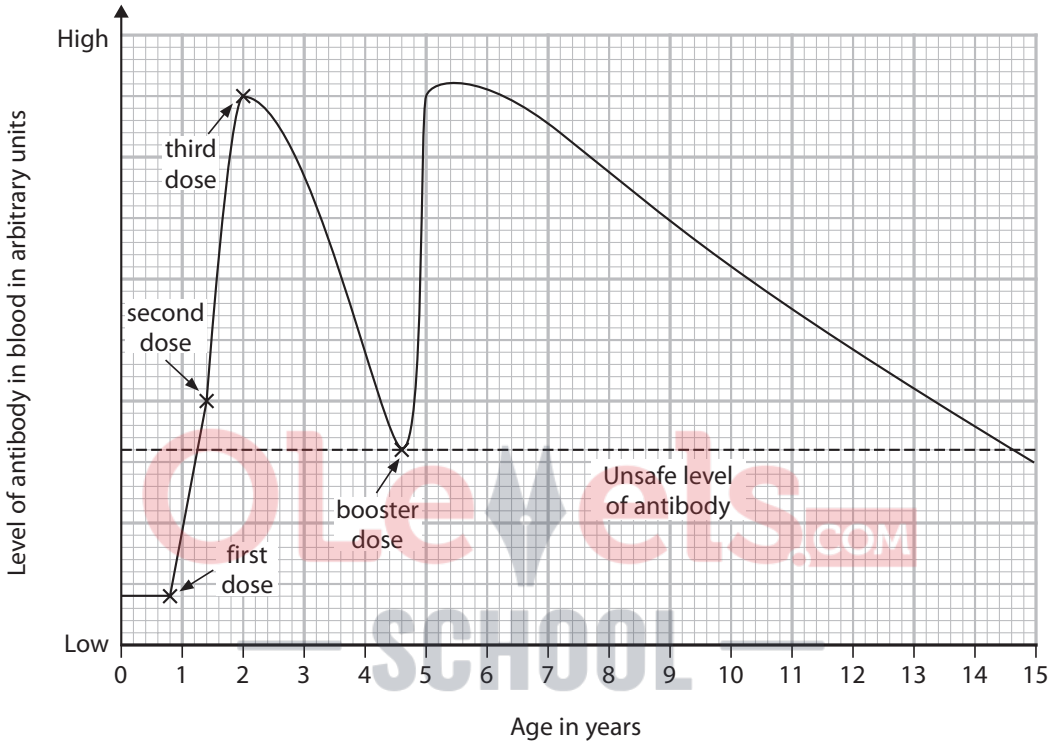
Further practice · Attempt part c only.

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time = ... years

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(2)

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.....

6. Question 3 · part c · 2 marks

November 2023 Paper 2

Further practice · Attempt part c only.

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(2)

1 .....

2 .....

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6. Question 3 · part c · 2 marks · Working and answer

Working and answer

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7. Question 8 · part a.vi · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.vi only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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7. Question 8 · part a.vi · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.vi only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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7. Question 8 · part a.vi · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.vi only.

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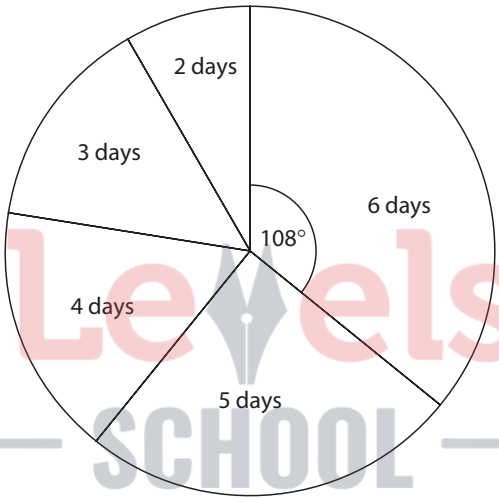
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(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.



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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

(3)

number of types of antibiotic = ...

7. Question 8 · part a.vi · 2 marks · Working and answer

Working and answer

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